CHANGE OF RING AND PULLBACK OF PRESHEAVES OF MODULES IS LEFT ADJOINT TO PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sheaves.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Tensor-restrict adjunction for presheaves of modules: Hom_{O}(O tensor_{p,f_pf_*O} f_pG, F) = Hom_{f_pf_*O}(f_pG, F_{f_pf_*O}). invoked at proof lines 3086-3087. Provides the first equality of the chain.
- Push-pull adjunction for presheaves of f_pf_*O-modules: Hom_{f_pf_*O}(f_pG, F) = Hom_{f_*O}(G, f_*F). invoked at proof lines 3088-3089. Provides the second equality of the chain.
- The adjunction unit and counit satisfy id_{f_*O} = f_*c_O composed with i_{f_*O}, making f_*(F_{f_pf_*O}) = f_*F as f_*O-modules. invoked at proof lines 3090-3096. Provides the third equality identifying the middle term with f_*F.

Source reference. Original lemma 008T: sources/stacks-sheaves/pinned-original-source.tex, source lines 2944–2944. Cited at statement lines 3064-3065 to name the pullback of module presheaves used in the tensor product; the construction is part of the asserted bijection.

Source reference. Original lemma 008S: sources/stacks-sheaves/pinned-original-source.tex, source lines 2910–2910. Cited at statement lines 3064-3065 to name the module structure on f_*F used on the right-hand side.

Source reference. Original lemma 006R: sources/stacks-sheaves/pinned-original-source.tex, source lines 692–692. Cited at proof lines 3086-3087 as the justification of the first equality of the chain; load-bearing inside the counted span.

Source reference. Original lemma 008U: sources/stacks-sheaves/pinned-original-source.tex, source lines 2999–2999. Cited at proof lines 3088-3089 as the justification of the second equality of the chain; load-bearing inside the counted span.

Source reference. Original lemma 008F: sources/stacks-sheaves/pinned-original-source.tex, source lines 2284–2284. Cited at proof lines 3091-3093 for the adjunction under which id_{f_*O} corresponds to c_O; load-bearing for the third equality.

Date: stacks-project:sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $f : X \to Y$ be a continuous map of topological spaces. Let $\mathcal{O}$ be a presheaf of rings on X. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$-modules. Let $\mathcal{G}$ be a presheaf of $f_*\mathcal{O}$-modules. Then*

$$\mathrm{Mor}_{PMod(\mathcal{O})}(\mathcal{O} \otimes_{p,f_pf_*\mathcal{O}} f_p\mathcal{G}, \mathcal{F}) = \mathrm{Mor}_{PMod(f_*\mathcal{O})}(\mathcal{G}, f_*\mathcal{F}).$$

Here we use Lemmas 008T and 008S, and we use the map $c_{\mathcal{O}} : f_pf_*\mathcal{O} \to \mathcal{O}$ in the definition of the tensor product.

Proof. This follows from the equalities

$$\begin{aligned} \mathrm{Mor}_{PMod(\mathcal{O})}(\mathcal{O} \otimes_{p,f_pf_*\mathcal{O}} f_p\mathcal{G}, \mathcal{F}) &= \mathrm{Mor}_{PMod(f_pf_*\mathcal{O})}(f_p\mathcal{G}, \mathcal{F}_{f_pf_*\mathcal{O}}) \\ &= \mathrm{Mor}_{PMod(f_*\mathcal{O})}(\mathcal{G}, f_*(\mathcal{F}_{f_pf_*\mathcal{O}})) \\ &= \mathrm{Mor}_{PMod(f_*\mathcal{O})}(\mathcal{G}, f_*\mathcal{F}). \end{aligned}$$

The first equality is Lemma 006R. The second equality is Lemma 008U. The third equality is given by the equality $f_*(\mathcal{F}_{f_pf_*\mathcal{O}}) = f_*\mathcal{F}$ of abelian sheaves which is $f_*\mathcal{O}$-linear. Namely, $\mathrm{id}_{f_*\mathcal{O}}$ corresponds to $c_{\mathcal{O}}$ under the adjunction described in the proof of Lemma 008F and thus $\mathrm{id}_{f_*\mathcal{O}} = f_*c_{\mathcal{O}} \circ i_{f_*\mathcal{O}}$. □